Q1.

A rocket was fired above a seaside town to call out the lifeboat crew.
The rocket exploded, giving out light and sound at the same time.

(a) Lisa was outside the town. She saw the flash of the rocket exploding and heard the bang.

(i) Which sentence is true?
Tick the correct box.

She heard the bang first. ☐

She saw the flash first. ☐

She heard the bang and saw the flash at the same time. ☐

1 mark

(ii) Give the reason for your answer.

.....

.....

1 mark

(b) Some people were nearer to the rocket than Lisa. How did the sound seem to them?

Tick the correct box.

It was quieter. ☐

It was louder. ☐

It was higher pitched. ☐

It was lower pitched. ☐

1 mark
Maximum 3 marks

Q2.

The table shows the speed at which sound travels through different materials.

material	physical state of the material	speed of sound in m/s
granite	solid	5400
nitrogen	gas	354
oil	liquid	1460
oxygen	gas	332
steel	solid	5980
water	liquid	1510

(a) Does sound travel fastest through solids or liquids or gases?

.....

1 mark

(b) Sound travels through air. Air is mostly nitrogen and oxygen.

Use the information in the table to suggest the speed of sound in a mixture of nitrogen and oxygen.

The speed of sound in this mixture is about …… m/s

1 mark

(c) Tom and Zoe make a string telephone.
They use two empty tin cans joined by a piece of string.



Tom and Zoe are 5 m apart.
Tom talks quietly into one tin can, and Zoe holds the other tin can to her ear.
If they keep the string tight, the sound of Tom's voice travels along the string.

(i) How does the sound travel along the string to Zoe?

.....

.....

1 mark

(ii) How does the tin can at Zoe's end make a sound?

.....

.....

1 mark
Maximum 4 marks

Mark schemes

Q1.

(a) (i) She saw the flash first. ✓

if more than one box is ticked award no mark

1

(ii) any **one** from

- light travels faster than sound

  accept 'the flash travelled faster than the bang'

- sound travels more slowly than light

  accept 'the bang travelled more slowly than the flash'
  *do **not** accept 'the flash **or** light travels very fast'*
  *do **not** accept 'sound takes time to travel'*

1

(b) It was louder. ✓

if more than one box is ticked, award no mark

1

[3]

Q2.

(a) solids

accept 'solid'

1 (L3)

(b) any number between 333 and 353 inclusive

1 (L4)

(c) (i) as waves **or** vibrations

*accept 'the string vibrates' **or** 'it vibrates'*

1 (L4)

(ii) it vibrates

accept 'it makes the air vibrate'

1 (L4)

[4]